

EXERCISE 3

Create a 'Hello World' Application

ABAP Development Tools (ADT) • ABAP Cloud

COURSE VALUES

Group number: 9309

Package: ZS4D400_9309

Class: ZCL_9309_HELLO_WORLD

Interface: IF_OO_ADT_CLASSRUN

Transport request: L02K910107

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This document contains only development information and my exercise screenshots. Do not add service URLs, passwords, browser login information, or other access credentials to this document or to public repositories.

1. Exercise Objective

In this exercise, you create my first executable ABAP Cloud class. The class implements IF_OO_ADT_CLASSRUN so that Eclipse ADT can execute it as a console application. The program then writes the text 'Hello World' to the ABAP Console.

What are we learning?

- How to create a global ABAP class inside my own package.
- How to implement an interface.
- What IF_OO_ADT_CLASSRUN does.
- What the MAIN method is used for.
- How out->write() sends output to the Eclipse ABAP Console.
- How to activate an ABAP class.
- How to execute a console application from ADT.

2. Task 1 — Create the ABAP Class

Create a new global class in package ZS4D400_9309. The class name follows the course naming convention:

```
ZCL_9309_HELLO_WORLD
```

Step-by-step

1. Open the class wizard. Choose File → New → ABAP Class.

2. Select the package. Enter/select ZS4D400_9309 in the Package field.

3. Enter the class name. Enter ZCL_9309_HELLO_WORLD.

4. **Enter a description.** For example: Hello World Application.
5. **Add the interface.** Choose Add... next to Interfaces.
6. **Search for the interface.** Enter IF_OO_ADT_CLASSRUN in the filter field.
7. **Select the interface.** Double-click IF_OO_ADT_CLASSRUN in the result list.
8. **Continue.** Choose Next.
9. **Assign the transport.** Select the transport request you created earlier: L02K910107.
10. **Create the class.** Choose Finish.

3. Understand the Generated Class

After creating the class, ADT generates a class definition similar to this:

```
CLASS zcl_9309_hello_world DEFINITION
  PUBLIC
  FINAL
  CREATE PUBLIC .

  PUBLIC SECTION.

      INTERFACES if_oo_adt_classrun.

  PROTECTED SECTION.
  PRIVATE SECTION.
ENDCLASS.
```

The important line for this exercise is:

```
INTERFACES if_oo_adt_classrun.
```

This tells ABAP that my class implements the IF_OO_ADT_CLASSRUN interface.

Simple meaning of interface

An interface is a contract. It tells a class which required behavior/methods it must provide. Here, the interface gives my class the MAIN method that ADT can use to run the class.

4. What Does IF_OO_ADT_CLASSRUN Mean?

You do not need to memorize the technical abbreviation character by character. For this course, remember the practical meaning:

```
IF_OO_ADT_CLASSRUN
```

It is an SAP-provided interface used to make a class executable as a console application in ABAP Development Tools.

Breakdown

Part	Meaning
IF	Interface

OO	Object-Oriented
ADT	ABAP Development Tools
CLASSRUN	The class can be run by ADT as a console application

5. Task 2 — Write Hello World

Inside the generated MAIN method, enter:

```
METHOD if_oo_adt_classrun~main.
    out->write( 'Hello World' ).
ENDMETHOD.
```

My completed class should look like:

```
CLASS zcl_9309_hello_world DEFINITION
    PUBLIC
    FINAL
    CREATE PUBLIC .

    PUBLIC SECTION.

        INTERFACES if_oo_adt_classrun.

    PROTECTED SECTION.
    PRIVATE SECTION.
ENDCLASS.

CLASS zcl_9309_hello_world IMPLEMENTATION.

    METHOD if_oo_adt_classrun~main.
        out->write( 'Hello World' ).
    ENDMETHOD.

ENDCLASS.
```

6. Understand IF_OO_ADT_CLASSRUN~MAIN

The ~ symbol connects the interface name with the method implemented from that interface.

```
IF_OO_ADT_CLASSRUN~MAIN
```

Read it as: 'the MAIN method belonging to the IF_OO_ADT_CLASSRUN interface.'

Simple execution flow

Press F9 → ADT runs the class → MAIN method is executed → out->write() sends text → ABAP Console displays Hello World.

7. What Does out->write() Mean?

```
out->write( 'Hello World' ).
```

This calls the write method on the output object supplied to the MAIN method.

Code	Simple meaning
out	The output object provided by the class-run framework.
->	Object access operator; it accesses a method or attribute of an object.
write	The method that writes data to the console output.
'Hello World'	The text being sent to the console.

8. Screenshot — Completed Class and Console

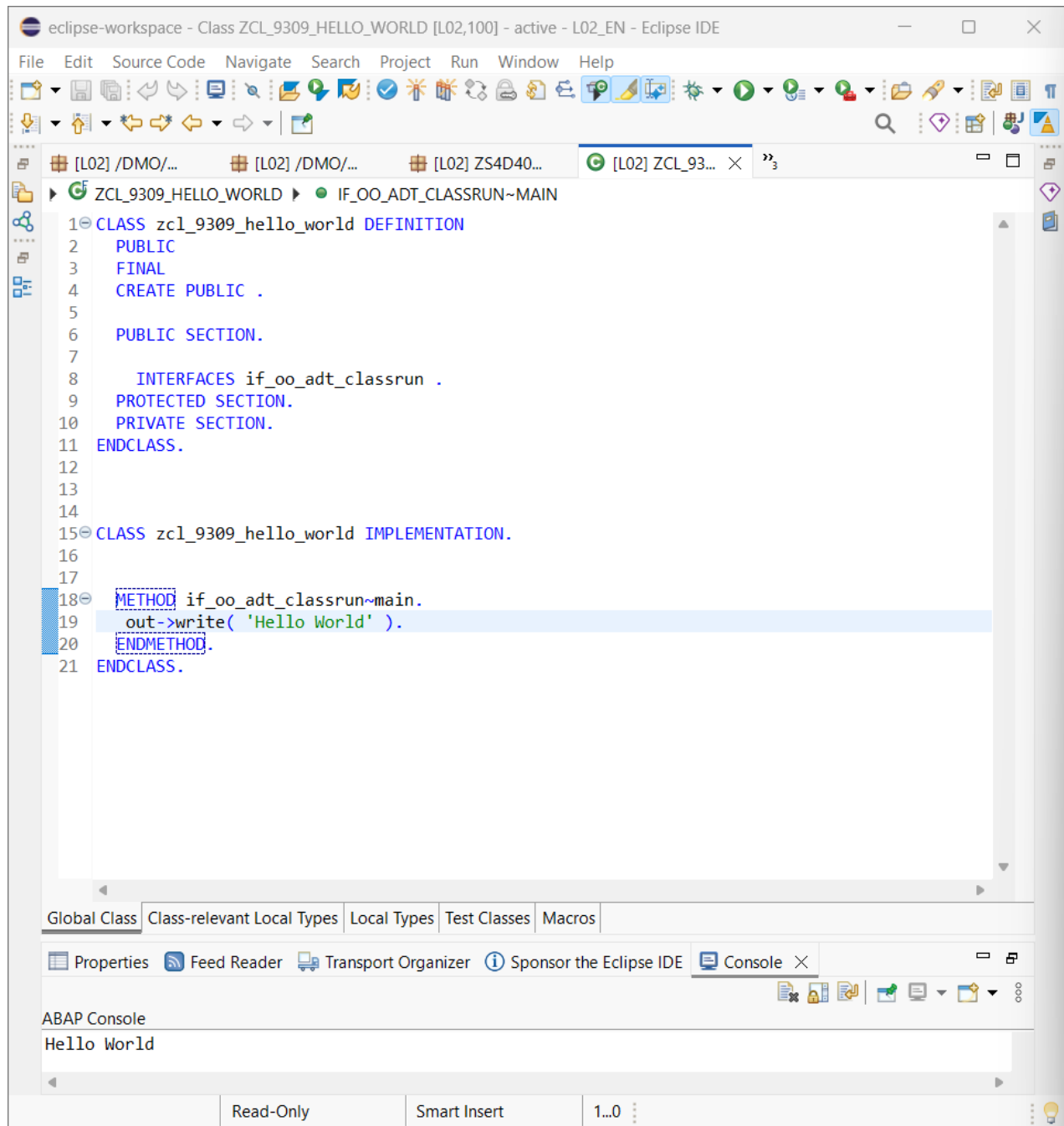


Figure 1 — My completed `ZCL_9309_HELLO_WORLD` class implementing `IF_OO_ADT_CLASSRUN` and displaying `Hello World` in the ABAP Console.

9. Task 3 — Activate the Class

ABAP source code must be activated before it can be executed. Activation checks and generates the active version of the development object.

- Save my changes if required.
- Press Ctrl + F3.
- Wait for the activation to finish.
- Check that there are no activation errors.

Activate vs Run

Activate does NOT mean execute. Activation makes the class's current source version active. Running the class actually executes the MAIN method.

10. Task 4 — Execute the Console Application

Run the class using:

F9

ADT executes the IF_OO_ADT_CLASSRUN-MAIN method. The result should appear in the ABAP Console.

Expected output

Hello World

If the Console is not visible

- Choose Window → Show View → Other...
- Search for Console.
- Double-click Console.
- Run the class again with F9.

11. Technical Vocabulary

Term	Simple meaning
Class	A blueprint containing behavior and data.
Global Class	A class stored as a repository development object and reusable from other ABAP objects.
Interface	A contract defining behavior that a class agrees to implement.
Method	A block of executable logic belonging to a class.
MAIN method	The entry point used by IF_OO_ADT_CLASSRUN when the class is executed as a console app.
Implementation	The actual executable code of the class/method.
Object	A runtime instance of a class.
ADT	ABAP Development Tools in Eclipse.
Console Application	A program whose output is displayed as text in a console rather than a graphical UI.
Activation	Makes the current ABAP source version active and checks it for errors.

Transport Request	Records development changes for transport/management in the SAP landscape.
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12. Why This Exercise Is Important in a Company

This looks trivial because the output is only 'Hello World'. The real purpose is not the message. It teaches the basic execution model you will use repeatedly in ABAP Cloud.

- You learn how ABAP classes are created and stored in packages.
- You learn how interfaces define required behavior.
- You learn how an executable class gets an entry point.
- You learn the ADT activation/run cycle.
- You learn how to test code without creating a full UI application.
- The same class-run pattern is useful for small tests, demonstrations, data-processing exercises, and learning ABAP syntax.

Real-company mindset

An interviewer is unlikely to care that you can print Hello World. They care whether you understand what happened: class → interface → MAIN entry point → executable code → console output, and whether you know how to activate and test it.

13. Common Problems and Solutions

Problem	What to check
F9 runs and disappears quickly	Make sure the class is activated and check the Console view. If necessary, open Console manually.
No MAIN method	Check that IF_OO_ADT_CLASSRUN was added to the class.
Activation error	Read the Problems view and fix the syntax/error before running.
No output	Check that out->write('Hello World'). is inside IF_OO_ADT_CLASSRUN~MAIN.
Wrong package	Verify the class belongs to ZS4D400_9309.
Transport prompt appears	Select the transport request assigned to my exercise, rather than creating unnecessary requests.

14. Interview Questions & Answers

1. What is IF_OO_ADT_CLASSRUN?

It is an SAP interface that allows an ABAP class to be executed as a console application from ADT.

2. Why do we implement IF_OO_ADT_CLASSRUN?

Because ADT needs the class-run interface to know which MAIN method should be used as the entry point when we execute the class.

3. What is an interface?

An interface is a contract. A class that implements it must provide the required interface methods.

4. What is the purpose of the MAIN method?

It is the entry point executed by the class-run framework when the console application starts.

5. What does the ~ symbol mean in IF_OO_ADT_CLASSRUN~MAIN?

It connects the interface name with the method being implemented. It means the MAIN method from that interface.

6. What does out->write() do?

It calls the write method of the output object and sends text/data to the ABAP Console.

7. What is the difference between activation and execution?

Activation makes the source version active and checks it. Execution actually runs the program.

8. Why is the class global?

A global class is stored in the ABAP repository and can be reused/referenced by other development objects.

9. What happens when F9 is pressed?

ADT executes the class as a console application, invoking IF_OO_ADT_CLASSRUN~MAIN.

10. Why do we use a transport request?

To record and manage the development change so it can be handled within the SAP development/transport process.

15. What We Performed — Final Summary

- Created ZCL_9309_HELLO_WORLD in package ZS4D400_9309.
- Implemented IF_OO_ADT_CLASSRUN.
- Used the interface's MAIN method as the executable entry point.
- Added out->write('Hello World').
- Activated the class using Ctrl + F3.
- Executed the class using F9.
- Verified Hello World in the ABAP Console.

One-line understanding

We created an ABAP Cloud class that implements IF_OO_ADT_CLASSRUN, put our program logic inside its MAIN method, and used out->write() to display the result in the Eclipse ABAP Console.